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SEPARATION AND PURIFICATION MODULE AND AUTOMATED MATERIAL REACTION SYSTEM INCLUDING THE SAME

Abstract

An automated material reaction system includes a material storage, which stores store at least one material, a dispensing module, which dispenses, to a reaction container, the at least one material stored in the material storage, a reaction module, which causes a reaction of the at least one material dispensed to the reaction container, a purification module, which extracts a target product from a product produced by the reaction module and purifies the target product, and a transport device, which transports the target product to the dispensing module such that the target product is used as a reactant.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based on and claims priority from Korean Patent Application No. 10-2024-0032764, filed on Mar. 7, 2024, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] The disclosure relates to a separation and purification module and an automated material reaction system including the separation and purification module.

2. Description of the Related Art

[0003] Research is being conducted on systems that automatically perform organic synthesis based on flow chemistry. However, in related art systems, user intervention was inevitably required in some parts of the synthesis recipe generation process to implement automation. In other words, in related art systems, user intervention was inevitable, limiting the capability of the system for developing materials in multiple steps. Therefore, there is a demand for an automated material reaction system in which the entire process may be performed automatically, allowing materials to react in multiple steps.

SUMMARY

[0004] One or more embodiments may address at least the problems and/or disadvantages described above, and other disadvantages not described above. Also, the embodiments are not required to overcome and may not overcome any of the problems and disadvantages described above.

[0005] According to an aspect of the disclosure, there is provided an automated material reaction system including: a material storage configured to store at least one material; a dispensing module configured to dispense, to a reaction container, the at least one material stored in the material storage; a reaction module configured to output a first product based on a reaction based on the at least one material dispensed to the reaction container; a separation and purification module configured to: extract a second product from the product output by the reaction module, and purify the second product; and a transport device configured to transport the second product to the dispensing module to be input as a reactant.

[0006] The transport device may be further configured to automatically transfer a material to be transferred between the material storage, the dispensing module, the reaction module, and the separation and purification module.

[0007] The separation and purification module may include: a work-up module configured to separate the second product from the first product output by the reaction module; and a purification module configured to purify the second product from a processing material of the work-up module.

[0008] The work-up module may include a filter or an evaporator.

[0009] The work-up module may further include a solvent supplier configured to supply a solvent

to the filter or the evaporator.
[0010] The purification module may include: a liquid chromatography device configured to separate the second product from the processing material processed of the work-up module; and a fraction collector configured to fractionize the second product separated by the liquid chromatography device.
[0011] The purification module may further include an evaporator configured to vaporize a solvent from the second product fractionized by the fraction collector.
[0012] The purification module may further include a weighing machine configured to measure a weight of the second product processed by the evaporator.
[0013] The purification module may further include a solvent supplier configured to supply a solvent to the evaporator, and the purification module may be further configured to supply the material processed by the evaporator back to the liquid chromatography device.
[0014] Each of the work-up module and the purification module may include a capper configured to attach a cap to a container or remove the cap from the container.
[0015] A material in the work-up module and the purification module may be automatically transported by a sub-transport device.
[0016] The automated material reaction system may include a product storage configured to store the second product.
[0017] The automated material reaction system may include an analysis module configured to analyze the second product.
[0018] The automated material reaction system may include a preprocessing module configured to preprocess the second product produced by the reaction module.
[0019] The automated material reaction system may include an artificial intelligence device configured to control the automated material reaction system to operation in an automated manner.
[0020] According to an aspect of the disclosure, there is provided a separation and purification module applied to an automated material reaction system, the separation and purification module including: a work-up module configured to separate, from a first product, a second product produced by a reaction module; and a purification module configured to purify the second product from a processing material of the work-up module, wherein the work-up module includes a filter or a first evaporator, and wherein the purification module includes a liquid chromatography device and a fraction collector.
[0021] The purification module further may include: a second evaporator configured to vaporize a first solvent from the second product fractionized by the fraction collector; and a solvent supplier configured to supply a second solvent the second product processed by the evaporator, wherein the purification module is further configured to supply the second product back to the liquid chromatography device.
[0022] The purification module may further include a weighing machine configured to measure a weight of the second product processed by the evaporator.
[0023] Each of the work-up module and the purification module may further include a capper configured to attach or remove a cap to or from a container.
[0024] A material in the work-up module and the purification module may be automatically transported by a sub-transport device.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0025] The above and/or other aspects will be more apparent from descriptions of certain embodiments referring to the accompanying drawings, in which:
[0026] FIG. **1** is a block diagram schematically illustrating an automated material reaction system

according to an embodiment;
[0027] FIG. **2** is a block diagram schematically illustrating a separation and purification module according to an embodiment;
[0028] FIG. **3** is a flowchart of a purification process according to an embodiment;
[0029] FIG. **4** is a flowchart of a purification process according to an embodiment;
[0030] FIG. **5** is a block diagram of an operation of an automated material reaction system, according to an embodiment;
[0031] FIG. **6** is a block diagram illustrating processes of a single-step reaction optimization platform and a scale-up purification platform, according to an embodiment;
[0032] FIG. **7** is a flowchart of a scale-up purification process according to an embodiment; and
[0033] FIG. **8** is a flowchart of an operation of an automated material reaction system according to an embodiment.

DETAILED DESCRIPTION

[0034] Hereinafter, embodiments will be described in detail with reference to the accompanying drawings. However, various alterations and modifications may be made to the embodiments. Here, the embodiments are not meant to be limited by the descriptions of the disclosure. The embodiments should be understood to include all changes, equivalents, and replacements within the idea and the technical scope of the disclosure.
[0035] The terminology used herein is for the purpose of describing particular embodiments only and is not to be limiting of the embodiments. The singular forms “a”, “an”, and “the” include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises/comprising” and/or “includes/including” when used herein, specify the presence of stated features, integers, steps, operations, elements, components, or groups thereof but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, or groups thereof.
[0036] Unless otherwise defined, all terms including technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the embodiments belong. Terms, such as those defined in commonly used dictionaries, are to be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the disclosure, and are not to be interpreted in an idealized or overly formal sense unless expressly so defined herein.
[0037] Furthermore, when describing the embodiments with reference to the accompanying drawings, like reference numerals refer to like components and a repeated description related thereto will be omitted. In the description of embodiments, detailed description of well-known related structures or functions will be omitted when it is deemed that such description will cause ambiguous interpretation of the disclosure.
[0038] In addition, terms such as first, second, A, B, (a), (b), and the like may be used to describe components of the examples. These terms are used only for the purpose of discriminating one component from another component, and the nature, the sequences, or the orders of the components are not limited by the terms. When one component is described as being “connected,” “coupled,” or “attached” to another component, it should be understood that one component may be connected or attached directly to another component, and an intervening component may also be “connected,” “coupled,” or “attached” to the components.
[0039] The same name may be used to describe an element included in the embodiments described above and an element having a common function. Unless otherwise mentioned, the description on one embodiment may be applicable to other embodiments and thus, duplicated descriptions will be omitted for conciseness.
[0040] FIG. **1** is a block diagram schematically illustrating an automated material reaction system according to an embodiment.
[0041] Referring to FIG. **1**, according to an embodiment, an automated material reaction system

100 may be configured to automatically perform a process of causing a reaction of a material. For example, the automated material reaction system **100** may be configured to automatically perform multi-step reactions. For example, the automated material reaction system **100** may be configured to automatically perform consecutive multi-step reactions. For example, the automated material reaction system **100** may be configured to perform a first reaction such that a product resulting from the first reaction is used as a reactant of a second reaction. For example, the automated material reaction system **100** may be configured to perform the first reaction and then scale up and/or purify the product resulting from the first reaction such that the product may be used as a reactant of the second reaction. For example, the entire process of the automated material reaction system **100** may be automated, requiring no user intervention. For example, the automated material reaction system **100** may be automated and operated by artificial intelligence (e.g., an artificial intelligence device **190**). For example, the automated material reaction system **100** may autonomously synthesize predetermined materials to optimize a corresponding reaction recipe. For example, the automated material reaction system **100** may use the optimized reaction recipe to perform a scale-up reaction and obtain a target material through a separation and purification process. For example, the automated material reaction system **100** may automatically perform a process of inputting the material obtained through scale-up as a source of the subsequent reaction. Each process of the automated material reaction system **100** may be performed repeatedly and/or autonomously. For example, the automated material reaction system **100** may be utilized in the material development field and/or pharmaceutical field and may contribute to lab automation. For example, the automated material reaction system **100** may automatically perform organic synthesis based on flow chemistry. For example, the automated material reaction system **100** utilizing artificial intelligence technology may be used to autonomously improve an organic synthesis method or suggest a new organic synthesis method.

[0042] In an embodiment, the automated material reaction system **100** may include a material storage **110**, a dispensing module **120**, a reaction module **130**, a preprocessing module **140**, a separation and purification module **150**, a product storage **160**, an analysis module **170**, a transport device **180**, and/or the artificial intelligence device **190**. However, these are only examples, and components of the automated material reaction system **100** are not limited thereto. As such, one or more components may be added and/or removed from the automated material reaction system **100**. According to an embodiment, the automated material reaction system **100** may include a controller or a control circuit. In some embodiments, the controller or the control circuit may be may be provided in the components or modules of the automated material reaction system **100**.

[0043] For example, the controller may be a memory and a processor. For example, the memory as a storage for storing data may store, for example, various algorithms, various programs, and various data. The memory may store one or more instructions. The memory may include at least one of volatile memory and non-volatile memory. The non-volatile memory may include read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable and programmable ROM (EEPROM), flash memory, phase-change random access memory (PRAM), magnetic RAM (MRAM), or resistive RAM (RRAM). The volatile memory may include dynamic RAM (DRAM), static RAM (SRAM), synchronous DRAM (SDRAM), PRAM, MRAM, or RRAM. In an embodiment, the memory may include at least one of a hard disk drive (HDD), a solid state drive (SSD), compact flash (CF), secure digital (SD), micro-SD, mini-SD, extreme digital (xD), and a memory stick. In an embodiment, the memory **1100** may semi-permanently or temporarily store algorithms, programs, and one or more instructions executed by the processor.

[0044] The processor may control an overall operation of the automated material reaction system **100**. The processor may include one or more of a central processing unit (CPU), an application processor (AP), and a communication processor (CP). The processor may perform, for example, an operation or processing related to control and/or communication of at least one other component of the automated material reaction system **100**. For example, the processor may execute one or more instructions corresponding to one or more of the various algorithms, various programs.

[0045] In an embodiment, the material storage **110** may be a component for storing at least one material. A material stored in the material storage **110** may be a predetermined material used for a chemical reaction. For example, the material stored in the material storage **110** may include a reactant, a solvent, and/or a catalyst. The reactant may be a sample and/or a reagent. However, the material stored in the material storage **110** is not limited thereto. The material storage **110** may provide a designated environment for storing the material. For example, the material storage **110** may include a pantry, a refrigerator, a warmer, and/or a vacuum chamber. However, these are only examples, and the type of the material storage **110** is not limited thereto.

[0046] In an embodiment, the dispensing module **120** may be a component for dispensing at least one material stored in the material storage **110** to a reaction container. For example, the dispensing module **120** may include a module for dispensing a liquid material and/or a module for dispensing a solid material. For example, the dispensing module **120** may dispense a designated material to a designated container with a designated volume or weight according to an input command. For example, the dispensing module **120** may include a capper for attaching a cap to a container or removing the cap from the container. For example, the dispensing module **120** may include a robot arm for gripping and/or transporting the container. For example, the dispensing module **120** may include a device for measuring the volume or weight of the material. However, this is only an example, and the function and/or components of the dispensing module **120** are not limited thereto.

[0047] In an embodiment, the reaction module **130** may be a component for causing a reaction of at least one material dispensed to a reaction container. For example, the reaction module **130** may be configured to provide an environment required for a chemical reaction to occur. For example, the reaction module **130** may include a reaction room for providing a designated reaction condition. For example, the designated reaction condition may include, but is not limited to, temperature, pressure, and/or humidity, etc. For example, the reaction module **130** may include a mixer or a stirrer that may consistently mix a material. However, this is only an example, and the function and/or components of the reaction module **130** are not limited thereto.

[0048] In an embodiment, the preprocessing module **140** may be a component for preprocessing a product produced in the reaction module **130**. For example, the preprocessing module **140** may be a component for preprocessing a material before analyzing the material. For example, the preprocessing module **140** may undergo a process of purifying, cleaning, separating, concentrating, diluting, and/or extracting a material. However, this is only an example, and the function of the preprocessing module **140** is not limited thereto.

[0049] In an embodiment, the separation and purification module **150** may be a component for extracting a target product from a product produced by the reaction module **130** and purifying the target product. For example, the separation and purification module **150** may be configured to extract a target product or a target component from a mixture produced after the chemical reaction. For example, the separation and purification module **150** may be configured to extract only a target product from a product produced after a chemical reaction and increase purity. The separation and purification module **150** is described in detail below.

[0050] In an embodiment, the product storage **160** may be a component for storing a product (e.g., a target product). A material stored in the product storage **160** may be an intermediate product or a final product. However, a material stored in the product storage **160** is not limited thereto. The product storage **160** may provide a designated environment for storing the material. For example, the product storage **160** may provide a designated environment for storing the target product extracted by the separation and purification module **150**. For example, the product storage **160** may include a pantry, a refrigerator, a warmer, and/or a vacuum chamber. However, these are only examples, and the type of the product storage **160** is not limited thereto. For example, the product storage **160** may include a robot arm, a capper, and/or a pipette device. For example, the product storage **160** may include components configured to operate autonomously.

[0051] In an embodiment, the analysis module **170** may be a component for analyzing a product (e.g., a target product). For example, the analysis module **170** may perform liquid chromatography-mass spectrometry (LC-MS), ultraviolet-visible photoluminescence (UV-PL), and/or nuclear magnetic resonance (NMR) analyses. For example, the analysis module **170** may include a liquid chromatography device, a mass spectrometry device, an infrared-visible light spectroscopic device, and/or a nuclear magnetic resonance spectroscopic device. However, these are only examples, and the type of the analysis module **170** is not limited thereto.

[0052] In an embodiment, the transport device **180** may be a component for transporting a material and/or a container in the automated material reaction system **100**. For example, the transport device **180** may be a component for transporting a material and/or a container between and/or within each module. For example, the transport device **180** may transport a material and/or a container between the material storage **110**, the dispensing module **120**, the reaction module **130**, the preprocessing module **140**, the separation and purification module **150**, the product storage **160**, and the analysis module **170**. For example, the transport device **180** may transport a material and/or a container within each of the product storage **110**, the dispensing module **120**, the reaction module **130**, the preprocessing module **140**, the separation and purification module **150**, the product storage **160**, and the analysis module **170**. For example, the transport device **180** may include a robot arm, a picker, a gripper, and/or a transport rail. However, these are only examples, and the function and/or components of the transport device **180** are not limited thereto. For example, the transport of a material and/or a container not elaborated in this disclosure is understood to be performed by the transport device **180**.

[0053] In an embodiment, the artificial intelligence device **190** may be an artificial intelligence-based device. For example, the artificial intelligence device **190** may be a processor and/or a memory that may use artificial intelligence. For example, the processor may be a neural network processor. The neural network processor may generate a neural network model, may train or learn the neural network model, may perform an operation based on received input data, may generate an information signal based on the operation result, or may retrain or update the neural network model. The neural network processor may process an operation based on various types of networks such as a convolution neural network (CNN), a region with a convolution neural network (R-CNN), a region proposal network (RPN), a recurrent neural network (RNN), a stacking-based deep neural network (S-DNN), a state-space dynamic neural network (S-SDNN), a deconvolution network, a deep belief network (DBN), restricted Boltzman machine (RBM), a fully convolutional network, a long short-term memory (LSTM) network, and a classification network. However, the disclosure is not limited thereto, and various types of computational processing that simulates human neural networks may be performed. The neural network processor may be implemented as a neural network operation accelerator, a coprocessor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a graphics processing unit (GPU), a neural processing unit (NPU), a tensor processing unit (TPU), or a multi-processor system-on-chip (MPSoC). The neural network processor may include one or more processors to perform operations according to neural network models. In addition, the neural network processor may include separate memory for storing programs corresponding to the neural network models. The neural network processor may be referred to as a neural network processing device, a neural network integrated circuit, or a neural network processing unit (NPU).

[0054] The artificial intelligence device **190** may autonomously drive the automated material reaction system **100** using artificial intelligence. For example, procedures performed by the artificial intelligence device **190** are presented by dashed lines in FIG. **1**. For example, the artificial intelligence device **190** may receive an analysis result (e.g., the yield and/or purity of a product and/or a by-product) from the analysis module **170**. The artificial intelligence device **190** may use the analysis result to control the material storage **110**, the dispensing module **120**, the reaction module **130**, the preprocessing module **140**, the separation and purification module **150**, the product storage **160**, and/or the analysis module **170**. For example, the artificial intelligence device **190** may use the analysis result to adjust the type of material (e.g., a reagent) discharged from the material storage **110**. For example, the artificial intelligence device **190** may use the analysis result to adjust a dispensing condition and/or method of the dispensing module **120**. For example, the artificial intelligence device **190** may use the analysis result to adjust reagent dispensing volume and the like. For example, the artificial intelligence device **190** may use the analysis result to adjust a reaction condition and/or method of the reaction module **130**. For example, the artificial intelligence device **190** may use the analysis result to adjust the reaction temperature, pressure, time, and/or humidity of the reaction module **130**. For example, the artificial intelligence device **190** may use the analysis result to adjust a preprocessing condition and/or method of the preprocessing module **140**. For example, the artificial intelligence device **190** may use the analysis result to adjust a separation and purification condition and/or method of the separation and purification module **150**. For example, the artificial intelligence device **190** may use the analysis result to optimize a purification method. For example, the analysis result may include, but is not limited to, final purity, material evaluation result after purification, and/or purification method evaluation result. For example, the artificial intelligence device **190** may optimize the purification method by repeatedly performing a process of modifying the purification method and analyzing and evaluating the purification result obtained through the modified purification method. For example, the artificial intelligence device **190** may enable purification to be performed with the optimized purification method in the separation and purification module **150**. For example, the artificial intelligence device **190** may adjust a condition and/or a method of storing and/or processing a material stored in the material storage **160**. For example, the artificial intelligence device **190** may adjust the analysis conditions and/or methods performed by the analysis module **170**. However, this is only an example, and the function of the artificial intelligence device **190** is not limited thereto.

[0055] FIG. **2** is a block diagram schematically illustrating a separation and purification module according to an embodiment.

[0056] Referring to FIG. **2**, according to an embodiment, the separation and purification module **150** may include a module interface **151**, a work-up module **200**, and/or a purification module **300**. However, these are only examples, and components of the separation and purification module **150** are not limited thereto. As such, one or more components may be added and/or removed from the separation and purification module **150**.

[0057] In an embodiment, the module interface **151** may be a part in which the separation and purification module **150** interacts with another module. For example, the module interface **151** may receive a material from a transport device (e.g., the transport device **180** of FIG. **1**) or transfer a material to the transport device (e.g., the transport device **180** of FIG. **1**). For example, in the module interface **151**, a material may be transported from the transport device (e.g., the transport device **180** of FIG. **1**) to the work-up module **200**. For example, in the module interface **151**, a material may be transported from the purification module **300** to the transport device (e.g., the transport device **180** of FIG. **1**).

[0058] In an embodiment, the work-up module **200** may be a component for separating a target product from a product or a result produced by a reaction module (e.g., the reaction module **130** of FIG. **1**). The work-up module **200** may include a capper **210**, a filter **220**, an evaporator **230**, and/or a solvent supplier **240**. However, these are only examples, and components of the work-up module **200** are not limited thereto.

[0059] In an embodiment, the capper **210** may be a component for attaching a cap to a container or removing the cap from the container. For example, the capper **210** may separate a cap from a container transported to the work-up module **200**. For example, in a case in which the cap is separated by the capper **210**, dilution may quench a reaction. The container, with the cap removed, may be transported to the filter **220**.

[0060] In an embodiment, the filter **220** may be a component for filtering a target material to separate the target material from a transported material. For example, the filter **220** may include various filter cartridges. For example, the filter **220** may eliminate a catalyst, water, and/or inorganic matter from the transported material. However, this is only an example, and the material filtered by the filter **220** is not limited thereto. The material processed by the filter **220** may be transported to the evaporator **230**.

[0061] In an embodiment, the evaporator **230** may be a component for eliminating a solvent from a material. For example, the evaporator **230** may eliminate the entire solvent by vaporizing the solvent from the transported material. For example, the evaporator **230** may be a rotary evaporator. However, this is only an example, the type of the evaporator **230** is not limited thereto. The material processed by the evaporator **230** may be transported to the solvent supplier **240**.

[0062] In an embodiment, the solvent supplier **240** may supply a solvent to the material processed by the filter **220** and/or the evaporator **230**. For example, the solvent supplier **240** may supply a material with a new solvent of a known concentration. The supplied solvent may be mixed with the material by a sonicator and/or a shaker. For example, the solver supplier **240** may receive a resulting material from the evaporator **230** and mix the new solvent with the mixture. The material processed by the solvent supplier **240** may be transported to the purification module **300** (e.g., an input device **310** of the purification module **300**).

[0063] In an embodiment, the purification module **300** may be a component for purifying the target product from the material processed by the work-up module **200**. The purification module **300** may include the input device **310**, a liquid chromatography device **320**, a fraction collector **330**, an evaporator **340**, a solvent supplier **350**, a weighing machine **360**, and/or a capper **370**. However, these are only examples, and components of the purification module **300** are not limited thereto.

[0064] In an embodiment, the input device **310** may be a component for inputting a material to the liquid chromatography device **320**. For example, the input device **310** may input the material received from the work-up module **200** into the liquid chromatography device **320**. For example, the input device **310** may perform a cleaning operation on the material received from the work-up module **200** to prevent cross-contamination before the material is input into the liquid chromatography device **320**.

[0065] In an embodiment, the liquid chromatography device **320** may be a component for performing liquid chromatography. For example, the liquid chromatography device **320** may separate the target product from the material processed by the work-up module **200**. For example, the liquid chromatography device **320** may separate a material by following a designated protocol.

[0066] In an embodiment, the fraction collector **330** may be a component for dividing the material separated from the liquid chromatography device **320**. For example, the material separated from the liquid chromatography device **320** may be subsequently fractionized by the fraction collector **330**.

[0067] In an embodiment, the evaporator **340** may be a component for vaporizing a solvent from the material fractionized by the fraction collector **330**. For example, the evaporator **340** may include a first rotary evaporator **341**, a second rotary evaporator **342**, and/or a vial evaporator **343**.

[0068] In an embodiment, the first rotary evaporator **341** may be a component for eliminating a solvent from the fractionized material. For example, the first rotary evaporator **341** may eliminate the entire solvent from the material by vaporizing the solvent from the fractionized material. The material processed by the first rotary evaporator **341** may be transported to the solvent supplier **350**.

[0069] In an embodiment, the solvent supplier **350** may be a component for supplying a solvent to the material processed by the first rotary evaporator **341** to supply the material processed by the first rotary evaporator **341** back to the liquid chromatography device **320**. For example, the solvent supplier **350** may supply the material with a new solvent of a known concentration. For example, the new solvent may be same or different from the solvent supplied by the solvent supplier of the work-up module **240**. The supplied solvent may be mixed with the material by a sonicator and/or a shaker. The first rotary evaporator **341** and the solvent supplier **350** may be components for recycling preprocessing. The material processed by the solvent supplier **350** may be transported to the input device **310** and then supplied back to the liquid chromatography device **320**. According to an embodiment, the process in which the material processed by the solvent supplier **350** may be transported back to the input device **310** and then supplied back to the liquid chromatography device **320** may be repeated a designated number of times.

[0070] In an embodiment, the second rotary evaporator **342** may be a component for eliminating the solvent from the fractionized material. For example, the second rotary evaporator **342** may eliminate the entire solvent from the material by vaporizing the solvent from the fractionized material. The material processed by the second rotary evaporator **342** may be transported to the vial evaporator **343**. According to an embodiment, the first rotary evaporator **341** and the second rotary evaporator **342** may have the same components.

[0071] In an embodiment, the vial evaporator **343** may be a component for inputting the material transported from the second rotary evaporator **342** into a vial and eliminating the solvent. For example, in the vial evaporator **343**, a designated amount of solvent may be input into an evaporating container to dilute and dissolve the material. The vial evaporator **343** may vaporize the entire solvent. The material processed by the vial evaporator **343** may be transported to the weighing machine **360**.

[0072] In an embodiment, the weighing machine **360** may be a component for measuring the weight of a material processed by an evaporator (e.g., the vial evaporator **343**). For example, the weighing machine **360** may measure the weight of a vial to measure the amount of the obtained material.

[0073] In an embodiment, the capper **370** may be a component for attaching or removing a cap to or from a container. For example, the capper **370** may couple a cap to a container transported from the weighing machine **360**. The container with the cap closed may be transported to the module interface **151** and then transported to a product storage (e.g., the product storage **160** of FIG. **1**), a dispensing module (e.g., the dispensing module **120** of FIG. **1**), and/or an analysis module (e.g., the analysis module **170** of FIG. **1**).

[0074] In an embodiment, the separation and purification module **150** may be configured such that each process of the separation and purification module **150** is automatically performed. For example, a material may be automatically transported in the work-up module **200** and/or the purification module **300** by a sub-transport device. For example, the sub-transport device may include a robot arm, a picker, a gripper, and/or a transport rail. The separation and purification module **150** may be automated and operated by artificial intelligence (e.g., the artificial intelligence device **190**). For example, the entire process of the separation and purification module **150** may be automated without a user's intervention such that the separation and purification module **150** autonomously separates and/or purifies a predetermined material.

[0075] Referring to FIG. **1**, in an embodiment, in the automated material reaction system **100**, a single reaction procedure and/or a multi-step reaction procedure may be performed.

[0076] Hereinafter, a single reaction procedure is described with reference to FIG. **1**. Referring to FIG. **1**, the single reaction procedure is represented by thin lines. In the single reaction procedure, at least one material stored in the material storage **110** may be dispensed to a reaction container (e.g., a vial) by the dispensing module **120**, and a chemical reaction may be performed by the reaction module **130**. A product produced by the reaction module **130** may be preprocessed by the preprocessing module **140** and then analyzed by the analysis module **170**. The preprocessing operation may include, but is not limited to, diluting or filtering the product produced by the reaction module **130**. For example, the analysis module **170** may detect a reaction status through liquid chromatography-mass spectrometry. For example, the analysis module **170** may analyze the amount of molecules, retention time, yield, and the presence or absence of a by-product and/or a starting material. For example, the analysis module **170** may infer a produced material from an analysis result. For example, while a reaction occurs in the reaction module **130**, a product may be periodically sampled such that the preprocessing and analysis described above are performed. However, this is only an example, and the single reaction procedure is not limited thereto.

[0077] Hereinafter, a multi-step reaction procedure is described with reference to FIG. **1**. The multi-step reaction procedure may be a procedure for maximizing yield or optimizing a reaction by repeatedly performing a single reaction procedure a plurality of times. Referring to FIG. **1**, the multi-step reaction procedure is represented by thick lines. In the multi-step reaction procedure, at least one material stored in the material storage **110** may be dispensed to a container (e.g., a vial) by the dispensing module **120**, and a chemical reaction may be performed by the reaction module **130**. The separation and purification module **150** may extract a target product from a product produced by the reaction module **130** and purify the target product. In an example, extracting the target product from the product produced by the reaction module **130** may including extracting a target component from a mixture of components in a result produced by the reaction module **130**. For example, the separation and purification module **150** may separate water, a catalyst, and/or inorganic matter from the product and separate a solute dissolved in a solvent through liquid chromatography. The target product, which is the result of the separation and purification module **150**, may be contained in a container (e.g., an intermediate vial) and stored in the product storage **160**. The transport device **180** may transport the target product to the dispensing module **120** such that the target product stored in the product storage **160** is used again as a reactant. For example, the product storage **160** may dispense the target product to a new container through pipetting, and the transport device **180** may transport the new container containing the target product to the dispensing module **120**. The dispensing module **120** may additionally dispense other materials to the container, as necessary. According to an embodiment, a chemical reaction for a subsequent step may be performed by the reaction module **130**, and the process described above may be consecutively and automatically repeated until the target product is obtained. The final target product may be transported to and stored in the product storage **160**. While the process described above is repeatedly performed, a target product at each step may be analyzed by the analysis module **170**. For example, the analysis module **170** may detect a reaction status through liquid chromatography-mass spectrometry. For example, the analysis module **170** may analyze the number of molecules, retention time, yield, and the presence or absence of a by-product, a starting material, and/or a final target product. For example, the analysis module **170** may infer a produced material from an analysis result. However, this is only an example, the multi-step reaction procedure is not limited thereto. For example, the target product separated and purified by the separation and purification module **150** may be transported direct to the dispensing module **120** and/or the analysis module **170** by the transported device **180** without passing through the product storage **160**. The multi-step reaction procedure described above may be autonomously performed by artificial intelligence (e.g., the artificial intelligence device **190**). With this structure, a multi-step reaction may be fully automated and implemented without user intervention.

[0078] FIG. **3** is a flowchart of a purification process according to an embodiment. As illustrated in FIG. **3**, a procedure performed by artificial intelligence is represented by dashed lines.

[0079] In an embodiment, the purification process as illustrated in FIG. **3** may be performed by the automated material reaction system **100** of FIG. **1**. Referring to FIGS. **1**, **2**, and **3**, a scale-up reaction may be performed by the reaction module **130** (S**10**). The result of the scale-up reaction may be transported to the purification module **150** (S**20**), and a reaction postprocessing process may be performed by the separation and purification module **150**. For example, the reaction postprocessing process may include a purification preprocessing process. The result of the reaction postprocessing process may be transported to the separation and purification module **150** and purified by the separation and purification module **150** (S**30**). In addition, the result of the scale-up reaction may be transported to the analysis module **170** and analyzed (S**40**). The artificial intelligence device **190** may optimize a purification method based on the analysis result of the analysis module **170**. For example, the artificial intelligence device **190** may optimize the purification method by repeatedly performing a process of modifying the purification method and analyzing and evaluating the purification result obtained through the modified purification method (S**50**). In the separation and purification module **150**, a separation and purification process may be performed using the optimized purification method (S**60**).

[0080] FIG. **4** is a flowchart of a purification process according to an embodiment. As illustrated in FIG. **4**, a procedure performed by artificial intelligence is represented by dashed lines.

[0081] In an embodiment, the purification process as illustrated in FIG. **4** may be performed by the automated material reaction system **100** of FIG. **1**. Referring to FIGS. **1**, **2**, and **3**, a scale-up reaction may be performed by the reaction module **130** (S**410**). The result of the scale-up reaction may be transported to the separation and purification module **150**, and a purification preprocessing process may be performed by the separation and purification module **150** (S**420**). For example, the purification preprocessing process may be performed by the work-up module **200**. The result of the purification preprocessing process may be purified by the purification module **300** (S**430**). In addition, the result of the scale-up reaction may be transported to and analyzed by the analysis module **170** (S**440**). The artificial intelligence device **190** may optimize a purification method and/or the purification preprocessing process based on the analysis result of the analysis module **170** (S**450**). For example, the purification preprocessing process may be based on output from the artificial intelligence device (S**460**). For example, the artificial intelligence device **190** may optimize the purification method by repeatedly performing a process of modifying the purification method and analyzing and evaluating the purification result obtained through the modified purification method (S**470**). In the purification module **300**, a purification process may be performed using the optimized purification method. In an embodiment, the purification result from the purification module **300** may be transported to and analyzed by the analysis module **170** (S**480**).

[0082] FIG. **5** is a block diagram of an operation of an automated material reaction system, according to an embodiment. As illustrated in FIG. **5**, a procedure performed by artificial intelligence is represented by a dashed line.

[0083] Referring to FIGS. **1**, **2**, and **5**, in an embodiment, the automated material reaction system **100** may be utilized as a platform for developing a material. For example, the automated material reaction system **100** may be utilized as a multi-step reaction platform. For example, the automated material reaction system **100** may include a single-step reaction optimization platform **101** and a scale-up purification platform **102**. The single-step reaction optimization platform **101** may be a platform for optimizing a single-step reaction. For example, the single-step reaction optimization platform **101** may be a platform for performing the operations connected by thin solid lines in FIG. **1**. For example, the single-step reaction optimization platform **101** may include the material storage **110**, the dispensing module **120**, the reaction module **130**, the preprocessing module **140**, the analysis module **170**, and/or the artificial intelligence device **190**. However, these are only examples, and components of the single-step reaction optimization platform **101** are not limited thereto. In the single-step reaction optimization platform **101**, an optimal synthesis method may be obtained as a result. For example, the optimal synthesis method may be obtained by artificial intelligence. The obtained optimal synthesis method may be applied to the scale-up purification platform **102**. For example, the scale-up purification platform **102** may be a platform for performing the steps connected by thick solid lines. For example, the scale-up purification platform **102** may include the material storage **110**, the dispensing module **120**, the separation and purification module **150**, the product storage **160**, the analysis module **170**, and/or the artificial intelligence device **190**. However, these are only examples, and components of the scale-up purification platform **102** are not limited thereto. For example, the scale-up purification platform **102** may scale up and/or purify a result by repeating a process. The scale-up purification platform

102 may produce great amounts of reagents (e.g., materials) using the optimal synthesis method. The result of the scale-up purification platform **102** may be used again as a source of the scale-up purification platform **102** or used as a source of the single-step reaction optimization platform **101**. Through such repetitive processes, a multi-step reaction may be conducted.

[0084] FIG. **6** is a block diagram illustrating processes of a single-step reaction optimization platform and a scale-up purification platform, according to an embodiment.

[0085] Referring to FIG. **6**, the single-step reaction optimization platform **101** may perform a low-volume reaction. For example, the low-volume reaction may be a single-step reaction. The low-volume reaction may mean that a sample evaluation timing is lower than a reference value and/or a reaction volume is lower than a reference value. However, the disclosure is not limited thereto. The scale-up purification platform **102** may perform a scale-up reaction and/or product purification. For example, the scale-up reaction may be a single-step reaction and may differ from the low-volume reaction in sample evaluation timing and/or reaction volume. The reaction volume in the scale-up reaction may be larger than the reaction in the low-volume reaction of the single-step reaction optimization platform **101**. The sample evaluation timing in the scale-up reaction may be larger than the sample evaluation timing in the low-volume reaction of the single-step reaction optimization platform **101**. The product purification process may include purification method optimization and a purification performing process. For example, the purification performing process may be a process of purifying the product produced by the scale-up reaction. For example, as the purification performing process is repeatedly performed, the purification method may be optimized by artificial intelligence. For example, the purification performing process and the purification method optimization may be performed in parallel and combined at a predetermined point.

[0086] FIG. **7** is a flowchart of a scale-up purification process according to an embodiment. As illustrated in FIG. **7**, a procedure performed by artificial intelligence is represented by a dashed line.

[0087] In an embodiment, the scale-up purification process according to FIG. **7** may be performed by the automated material reaction system **100** of FIG. **1**. Referring to FIGS. **1**, **2**, and **7**, a scale-up reaction may be performed by the reaction module **130** and/or the separation and purification module **150** (S**710**). Purification method optimization (S**720**) and/or a work-up process (S**730**) may be performed on the result of the scale-up reaction. The purification method optimization and the work-up process may be performed in parallel. As the purification method optimization is performed by artificial intelligence, an optimal purification method may be obtained (S**740**). For example, the artificial intelligence device **190** may obtain the optimal purification method by adjusting recipes of experiment methods. After the work-up process is performed, purification may be performed (S**750**). In this case, the obtained optimal purification method may be used. As purification is performed, the final product may be produced.

[0088] FIG. **8** is a flowchart of an operation of an automated material reaction system, according to an embodiment. As illustrated in FIG. **8**, procedures performed by artificial intelligence are represented by a dashed line.

[0089] In an embodiment, referring to FIGS. **1**, **2**, and **8**, the operation of the automated material reaction system may include a scale-up reaction **401**, purification method optimization **402**, and/or a separation and purification reaction **403**. For example, the scale-up reaction **401**, the purification method optimization **402**, and the separation and purification reaction **403** may be performed sequentially and/or in parallel. For example, the purification method optimization **402** and the separation and purification method **403** may be performed in parallel and may combined into a single step at a time point at which the purification method optimization **402** is completed.

[0090] In an embodiment, in the scale-up reaction **401**, a reagent stored in the material storage **110** may be discharged, the reagent is dispensed by the dispensing module **120**, and then a reagent mixture may be produced. The reagent mixture may be transported to the reaction module **130**, and the scale-up reaction may be performed by the reaction module **130**. At least a portion of the reaction result may be aliquoted as a sample, and the aliquoted sample may be analyzed by the analysis module **170**. The analysis result (e.g., yield, purity, and/or the like) may be transferred to the artificial intelligence device **190**. The artificial intelligence device **190** may use the received analysis result to adjust the type of reagent used for a reaction at the reagent discharging step. The artificial intelligence device **190** may use the received analysis result and adjust reagent dispensing volume, a dispensing environment, and/or the like at the reagent dispensing step. The artificial intelligence device **190** may use the received analysis result to adjust a reaction temperature, a reaction time, and/or the like at the scale-up reaction step. The artificial intelligence device **190** may use the received analysis result to adjust an analysis evaluation method of the analysis module **170**. However, this is only an example, and detailed steps of the scale-up reaction **401** and/or an operation of the artificial intelligence device **190** are not limited thereto.

[0091] In an embodiment, in the purification method optimization **402**, at least a portion of the reaction result of the scale-up reaction may be aliquoted as a sample, and the aliquoted sample may be analyzed by the analysis module **170**. The analysis result (e.g., a result of the evaluation of purification conditions for a sample) may be transferred to the artificial intelligence device **190**. The artificial intelligence device **190** may modify the purification condition according to the analysis result, and analysis may be performed by the analysis module **170** under the modified purification condition. By repeating the processes described above, the artificial intelligence device **190** may optimize the purification method. However, this is only an example, and detailed steps of the purification method optimization **401** and/or the operation of the artificial intelligence device **190** are not limited thereto.

[0092] In an embodiment, in the separation and purification reaction **403**, the reaction result (e.g., the remaining portion excluding the aliquoted sample) of the scale-up reaction may be transported to the work-up module **200**, and purification preprocessing (reaction postprocessing) may be performed by the work-up module **200**. A purification preprocessing method may be modified by the artificial intelligence device **190**. The result of the purification preprocessing may be transported to the purification module **300** and purified by the purification module **300**. Purification may be performed by the purification module **300** using the optimal purification method obtained by the artificial intelligence device **190**. At least a portion of the purification result may be aliquoted as a sample, and the aliquoted sample may be transmitted to and analyzed by the analysis module **170**. The remaining of the purification result excluding the aliquoted sample may be transported to and stored in the product storage **160**. However, this is only an example, and detailed steps of the separation and purification reaction **403** and/or the operation of the artificial intelligence device **190** are not limited thereto.

[0093] As described above, although the embodiments have been described with reference to the limited drawings, a person skilled in the art may apply various technical modifications and variations based thereon. For example, suitable results may be achieved if the described techniques are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or supplemented by other components or their equivalents.

Claims

1. An automated material reaction system comprising: a material storage configured to store at least one material; a dispensing module configured to dispense, to a reaction container, the at least one material stored in the material storage; a reaction module configured to output a first product based on a reaction based on the at least one material dispensed to the reaction container; a separation and purification module configured to: extract a second product from the first product output by the reaction module, and purify the second product; and a transport device configured to transport the second product to the dispensing module to be input as a reactant.

2. The automated material reaction system of claim 1, wherein the transport device is further configured to automatically transfer a material to be transferred between the material storage, the dispensing module, the reaction module, and the separation and purification module.

3. The automated material reaction system of claim 1, wherein the separation and purification module comprises: a work-up module configured to separate the second product from the first product output by the reaction module; and a purification module configured to purify the second product from a processing material of the work-up module.

4. The automated material reaction system of claim 3, wherein the work-up module comprises a filter or an evaporator.

5. The automated material reaction system of claim 4, wherein the work-up module further comprises a solvent supplier configured to supply a solvent to the filter or the evaporator.

6. The automated material reaction system of claim 3, wherein the purification module comprises: a liquid chromatography device configured to separate the second product from the processing material processed of the work-up module; and a fraction collector configured to fractionize the second product separated by the liquid chromatography device.

7. The automated material reaction system of claim 6, wherein the purification module further comprises an evaporator configured to vaporize a solvent from the second product fractionized by the fraction collector.

8. The automated material reaction system of claim 7, wherein the purification module further comprises a weighing machine configured to measure a weight of the second product processed by the evaporator.

9. The automated material reaction system of claim 7, wherein the purification module further comprises a solvent supplier configured to supply a solvent to the evaporator, and wherein the purification module is further configured to supply the material processed by the evaporator back to the liquid chromatography device.

10. The automated material reaction system of claim 3, wherein each of the work-up module and the purification module comprises a capper configured to attach a cap to a container or remove the cap from the container.

11. The automated material reaction system of claim 3, wherein a material in the work-up module and the purification module is automatically transported by a sub-transport device.

12. The automated material reaction system of claim 1, further comprising: a product storage configured to store the second product.

13. The automated material reaction system of claim 1, further comprising: an analysis module configured to analyze the second product.

14. The automated material reaction system of claim 1, further comprising: a preprocessing module configured to preprocess the second product produced by the reaction module.

15. The automated material reaction system of claim 1, further comprising: an artificial intelligence device configured to control the automated material reaction system to operation in an automated manner.

16. A separation and purification module applied to an automated material reaction system, the separation and purification module comprising: a work-up module configured to separate, from a first product, a second product produced by a reaction module; and a purification module configured to purify the second product from a processing material of the work-up module, wherein the work-up module comprises a filter or a first evaporator, and wherein the purification module comprises a liquid chromatography device and a fraction collector.

17. The separation and purification module of claim 16, wherein the purification module further comprises: a second evaporator configured to vaporize a first solvent from the second product fractionized by the fraction collector; and a solvent supplier configured to supply a second solvent the second product processed by the second evaporator, wherein the purification module is further

configured to supply the second product back to the liquid chromatography device.
18. The separation and purification module of claim 17, wherein the purification module further comprises a weighing machine configured to measure a weight of the second product processed by the second evaporator.
19. The separation and purification module of claim 16, wherein each of the work-up module and the purification module further comprises a capper configured to attach or remove a cap to or from a container.
20. The separation and purification module of claim 16, wherein a material in the work-up module and the purification module is automatically transported by a sub-transport device.